



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
24-12-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TTAS, TOAS, TNAS, TRAS, TMAS, TIAS, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**BEWARE OF NEGATIVE MARKING****PART-1 : PHYSICS****SECTION-I (i) : (Maximum Marks: 12)**

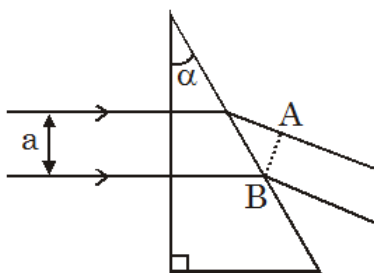
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A plane wavefront of monochromatic wave ($\lambda = 500$ nm) is falling on a glass prism ($n = \sqrt{2}$) and emerge as shown, out of it. The distance between incident rays $a = 2$ cm (Figure). Refracting angle prism, $\alpha = 30^\circ$. What is the difference of the phase between the light wave at point A and B after exit from the prism?



- (A) 0 (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Space for Rough Work

2. A parallel beam of light of intensity I_0 is incident on a glass plate, 25% of light is reflected by upper surface and 50% of light is reflected from lower surface. The ratio of maximum to minimum intensity in interference region of reflected rays is (Assume after reflection from lower surface light transmitted completely from upper surface).

(A) $\left[\frac{\frac{1}{2} + \sqrt{\frac{3}{8}}}{\frac{1}{2} - \sqrt{\frac{3}{8}}} \right]^2$ (B) $\left[\frac{\frac{1}{4} + \sqrt{\frac{3}{8}}}{\frac{1}{2} - \sqrt{\frac{3}{8}}} \right]^2$ (C) $\frac{5}{8}$ (D) $\frac{3}{5}$

3. A homogeneous ball (mass = m) of ideally black material at rest, is illuminated with a radiation having a set of photons (wavelength = λ) each with same momentum and same energy. The rate at which photons fall on the ball is n . The linear acceleration of the ball is

(A) $\frac{m\lambda}{nh}$ (B) $\frac{nh}{m\lambda}$ (C) $\frac{nh}{(2\pi)(m\lambda)}$ (D) $\frac{2\pi m\lambda}{nh}$

4. What is the essential distinction between X-rays and γ -rays ?

- (A) γ -rays have shorter wavelength than X-rays
 (B) γ -rays have less penetrating power than X-rays
 (C) γ -rays have greater ionizing powers than X-rays
 (D) γ -rays originate from within an atomic nucleus, X-rays from outside an atomic nucleus.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

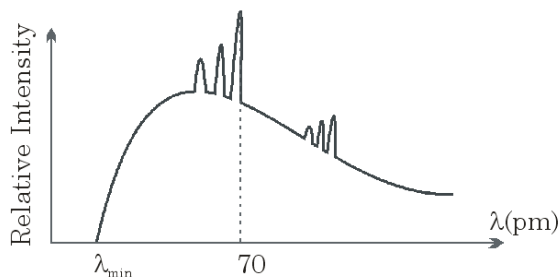
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

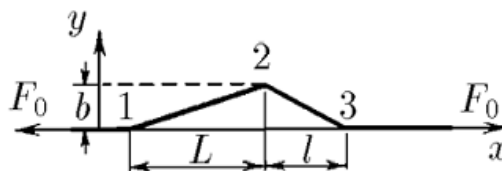
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5. When beats are produced by two progressive waves of nearly the same frequency, which one of the following (is/are) incorrect?
- (A) The particles vibrate simple harmonically, with the frequency equal to the difference in the component frequencies.
- (B) The amplitude of vibration at any point changes simple harmonically with a frequency equal to the difference in the frequencies of the two waves.
- (C) The frequency of beats depends upon the position, where the observer is
- (D) The frequency of beats changes as the time progresses.
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Space for Rough Work

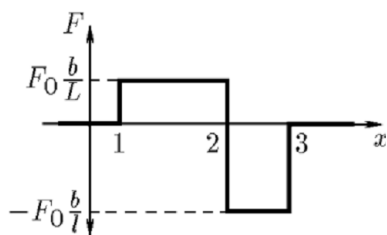
6. In the figure shown, X-rays are produced when 35 KeV electrons fall on a 'Mo' target. The accelerating potential is maintained at 35 keV but a instead of 'Mo' target, 'Ag' target ($Z = 47$) is placed. The K, L & M atomic X-ray levels for silver are 25.51, 3.56 & 0.53 KeV. Then for 'Ag' target (approx values are given) choose the correct option(s) :



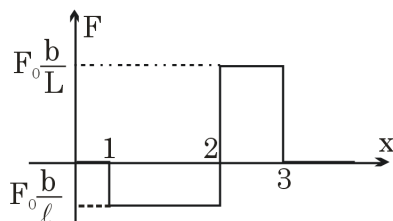
- (A) $\lambda_{\min} = 35.5 \text{ pm}$ (B) $\lambda_{K\alpha} = 56.63 \text{ pm}$ (C) $\lambda_{K\beta} = 44.76 \text{ pm}$ (D) none
7. Tensile force F_0 is applied to the ends of a horizontal string. A graph of transverse position of the string vs its X-coordinate is plotted (shown in figure below). And then a wave is sent through it.



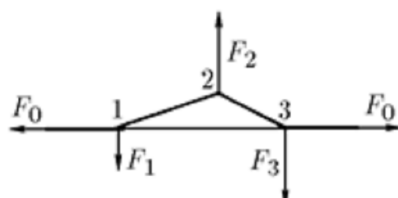
- (A) The plot the dependence of the transverse component of the string tension vs X – axis is:



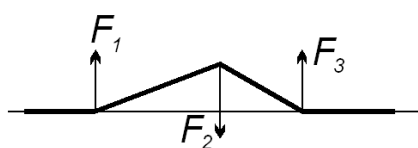
- (B) The plot the dependence of the transverse component of the string tension vs X – axis is:



- (C) The transverse forces can hold the string in this form is:

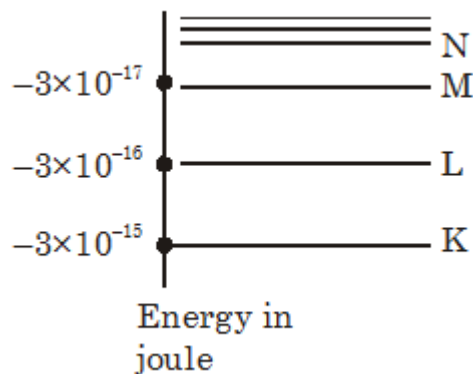


- (D) The transverse forces can hold the string in this form is:



Space for Rough Work

8. Simplified model of electron energy levels for a certain atom is shown in figure. The atom is bombard with fast moving electron. The impact of one of these electrons can cause the removal of electron from K-level, thus creating a vacancy in K-level. This vacancy in K-level is filled by an electron from L-level and the energy released in this transition can either appear as em- waves or may all be used to knock out an electron from M-level of the atom. Based on above information answer the following questions.



- (A) The minimum potential difference through which bombarding electron beam must be accelerated from rest to cause the ejection of electron from K-level is 18750 volt.
- (B) Wavelength of the electromagnetic waves emitted due to transition from L to K - level is 7.348×10^{-11} m.
- (C) The KE of the emitted electron from M - level is 267×10^{-17} J.
- (D) None of the above.

Space for Rough Work

9. A flat circular annular disc has a mass M , outer radius a & inner radius b . The various measured values are: $M = (0.300 \text{ kg} \pm 0.003 \text{ kg})$, $a = (4 \text{ cm} \pm 0.01 \text{ cm})$ & $b = (2 \text{ cm} \pm 0.01 \text{ cm})$:
- (A) Percentage error in the calculation of a and b are same.
 - (B) Percentage error in the calculation of M is 1%
 - (C) Percentage error in the calculation of moment of inertia of the disc about an axis passing through center and perpendicular to plane of disc is 1.6 %.
 - (D) Percentage error in the calculation of moment of inertia of the disc about an axis passing through center and perpendicular to plane of disc is 3.2 %.
10. Standing sound waves produced in a pipe that is 1.20 m long. For the first overtone determine the locations along the pipe (measured from the left end) of the displacement nodes (S_N) and the pressure nodes (P_N) if the pipe is
- (A) open at both ends, S_N at 0.3 m, 0.9 m
 - (B) open at both ends, P_N at 0 m, 0.6m, 1.2 m
 - (C) closed at the left end and open at the right end S_N at 0m, 0.8m,
 - (D) closed at the left end and open at the right end P_N at 0.4 m, 1.2 m.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

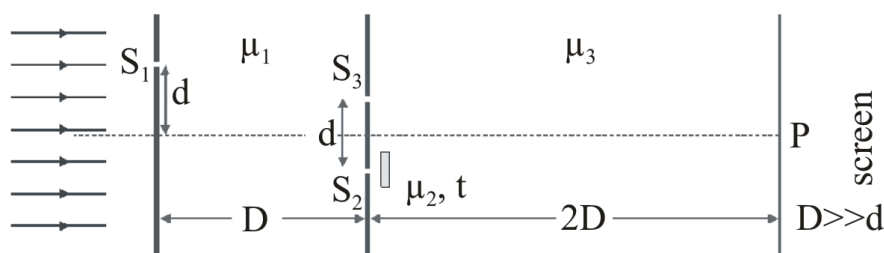
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

In the figure shown parallel light rays of wavelength λ (in air) fall on slit ' S_1 '. Then it reaches slits ' S_2 ' & ' S_3 ' through a medium of refractive index ' μ_1 '. Light then reaches the screen through medium of refractive index μ_3 . A thin transparent film of refractive index μ_2 and thickness ' t ' is placed in front of ' S_2 '. Point ' P ' is symmetrical w.r.t. ' S_2 ' and ' S_3 '. Using the values $d = 1\text{mm}$, $D = 1\text{m}$, $\mu_1 = 4/3$, $\mu_2 = 3/2$, $\mu_3 = 9/5$ and $t = (4/9) \times 10^{-5}\text{m}$.



1. Find the distance of central maxima from P.
2. If the film in front of S_2 is removed, then by what distance (in $\times 10^{-4}\text{m}$) will the central maxima shift?

Paragraph for Questions 3 and 4

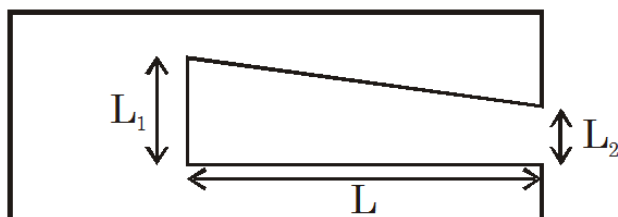
A student measures the density of a liquid five times and gets the results (all in gram/cm^3) 1.8, 2.0, 2.0, 1.9, and 1.8. He finds the mass of liquid to be $(152 \pm 1)\text{gm/cc}$.

3. Find the uncertainty in measuring the density of liquid (in gram/cm^3).
4. Find the error in determined value of volume of liquid (in percentage).

Space for Rough Work

Paragraph for Questions 5 and 6

A block of plastic having a thin air cavity (whose thickness is comparable to wavelength of light waves) is shown in figure. The thickness of air cavity (which can be considered as air wedge for interference pattern) is varying linearly from one end to other as shown.



Front View of Plastic Box

A beam of monochromatic light is incident normally from the top of plastic box. Some light is reflected back from top and some from bottom of cavity. The plastic layers above and below the cavity are having thickness much larger than wavelength of incident light. An observer when looking down from top sees an interference pattern consisting of 8 dark fringes and 7 bright fringes along the wedge. Take wavelength of incident light in air as λ_0 and Refractive index of plastic as μ . Assume that the thickness of the ends of air cavity are such that, formation of fringes takes place there.

5. Determine the difference $L_1 - L_2 (= \Delta L)$ in terms of λ_0 . If your answer is $\frac{\alpha \lambda_0}{\beta \mu}$ where α & β are least integer, then find the value of $\frac{\alpha}{\beta}$.
6. Determine the distance of 4th Dark fringe from left end of air cavity. If your answer is $\frac{\alpha L}{\beta}$ where α & β are least integer, then find the value of $(\beta - \alpha)$.

Space for Rough Work

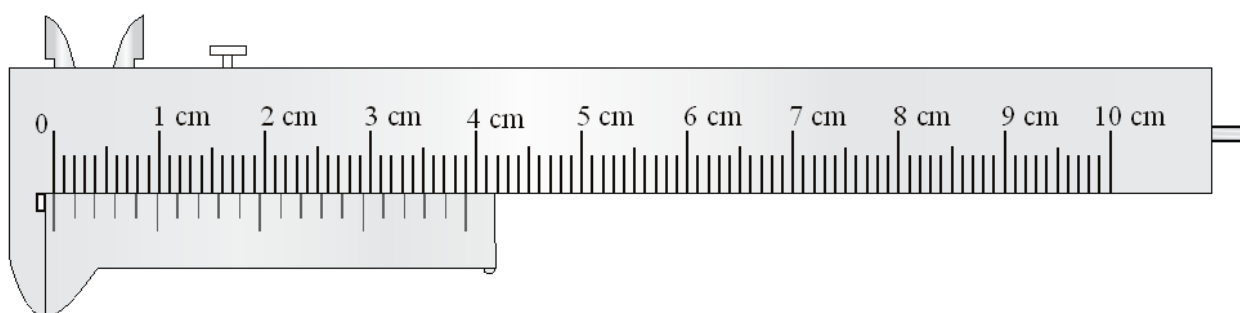
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

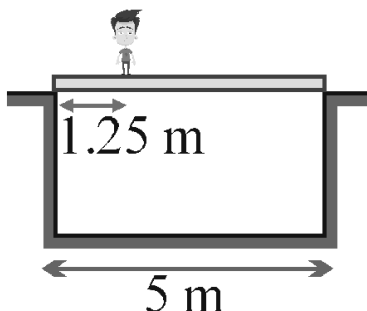
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A stationary nucleus of mass 25 u emits a photon of energy 3.05 MeV. Find the recoil energy of the nucleus (in eV) (Take 1 u = 931 MeV)
8. Find the LC of the vernier caliper (in μm) shown in the figure. The LC of the main scale is 1 mm.



9. A wooden flexible plank is placed over a pit which is 5m wide. A student begins to jump up and down at the center. Such that he jumps at a frequency of 2 Hz. The plank forms a standing wave pattern with minimum possible loops. Now, the student moves at 1.25 m from edge of the pit and jumps at this position such that this point has maximum amplitude. What is minimum frequency (in Hz) of his jumping ?



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Select the correct options for the following statements.

1. Cl_2O and ClO_2 are used as bleaching agents.
2. When HCl_{aq} react with Fe produces Ferrous chloride
3. OCl^- disproportionates to Cl^- & ClO_3^-
4. BrO_4^- is oxidized in acidic medium.

(A) 1,2,3 correct (B) 2,3,4 correct (C) 1,2,4 correct (D) 1,3,4 correct

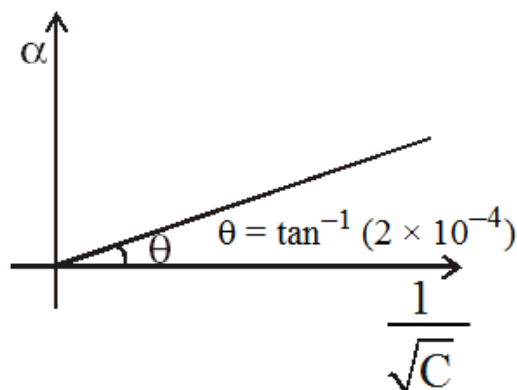
2. Which of the following reaction is incorrect here ?

- (A) $\text{Zn} + \text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
- (B) $\text{P}_4 + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{NaH}_2\text{PO}_2 + \text{PH}_3$
- (C) $\text{S}_8 + \text{NaOH} (\text{excess}) \rightarrow \text{Na}_2\text{S}_2\text{O}_3 + \text{Na}_2\text{S} + \text{H}_2\text{O}$
- (D) $\text{F}_2 + \text{NaOH} (\text{cold and dilute}) \rightarrow \text{NaF} + \text{NaOF} + \text{H}_2\text{O}$

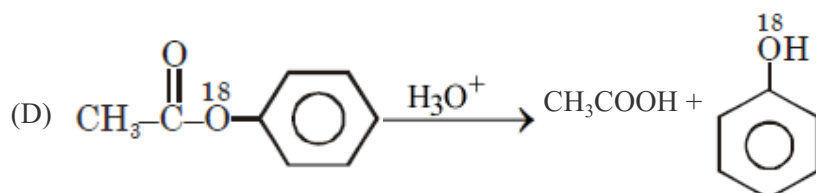
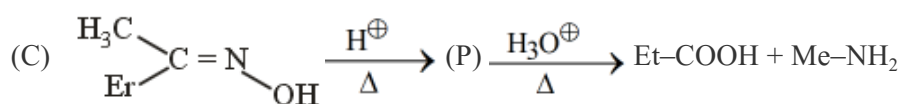
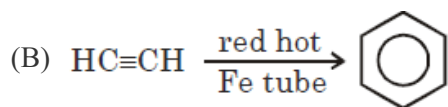
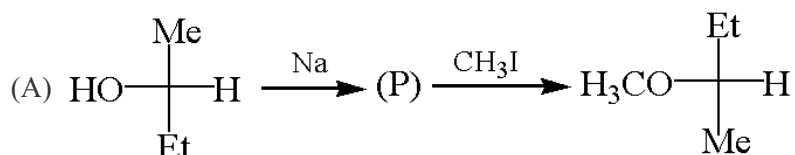
Space for Rough Work

3. For weak monobasic acid 'HA' the plot of α versus $\frac{1}{\sqrt{C}}$ is given below. What is the pH of a 0.01M solution of this acid 'HA'.

(α = degree of dissociation of acid, assume ' α ' $\ll 1$, C = concentration of acid, $\log 2 = 0.3$)



- (A) 4 (B) 3.7 (C) 4.7 (D) 4.3
4. Which of the following is incorrect sequence corresponding to final major product -



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

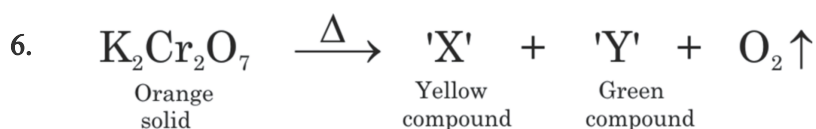
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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. When H_2O_2 reacts with $\text{Ce}(\text{SO}_4)_2$:-

- (A) Ce^{+4} is oxidised to higher oxidation state (B) Ce^{+3} is produced
(C) O_2 is produced (D) H_2 is evolved



Which of the following statement is **CORRECT**?

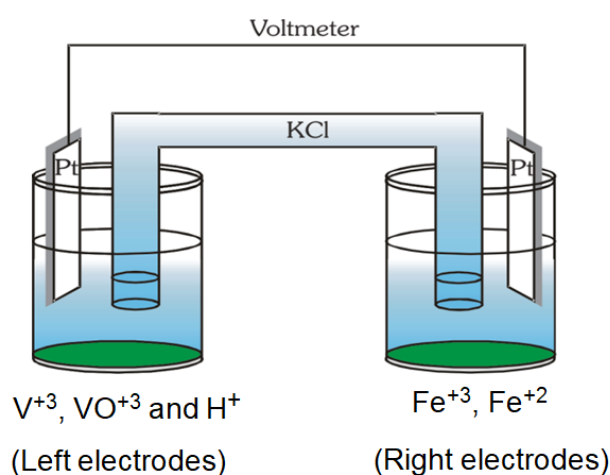
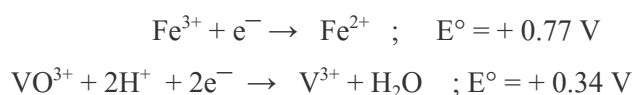
- (A) 'Y' is CrO_3
(B) Aqueous solution of 'X' with conc. H_2SO_4 is converted into orange solution
(C) When aqueous solution of 'X' is treated with H_2O_2 in acidic medium blue colour compound is obtained
(D) 'Y' is formed on thermal decomposition of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

Space for Rough Work

7. Select the correct statements(s) out of the following -

- (A) $[H^+]$ in 0.1 M $HCOOH$ solution is more than $[OH^-]$ in 0.1 M $HCOO^-$ solution at $25^\circ C$
 $[K_a(HCOOH) = 10^{-4}]$
- (B) If $[H^+]$ concentration decreases 100 times, pOH of solution decreases by 2 units
- (C) Buffer capacity of given buffer $[CH_3COOH (x \text{ molar}) + CH_3COO^- Na^+ (y \text{ molar})]$ increases to its maximum value if $\frac{y}{x}$ approaches zero.
- (D) Solubility of $Fe(OH)_3$ in 0.01M $Ca(OH)_2$ solution is $0.5 \times 10^{-32} M$ (Given $K_{sp} Fe(OH)_3 = 4 \times 10^{-38}$)

8. An electrochemical cell with inert platinum electrodes is shown below at $25^\circ C$. The half cell reactions are :

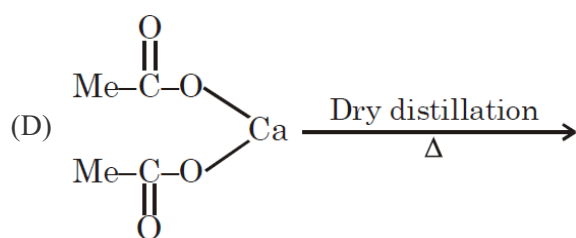
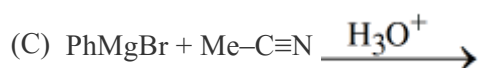
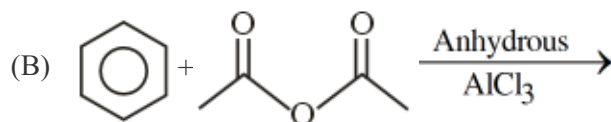


Select the statement(s) which is/are correct about this cell - $\left(\text{Take } \frac{2.303RT}{F} = 0.06 \right)$

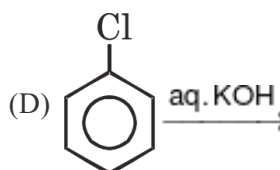
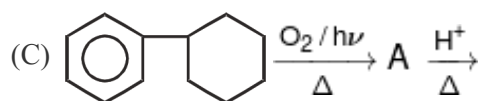
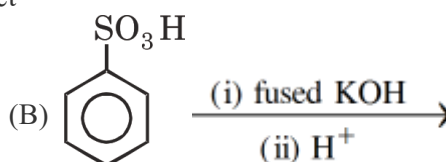
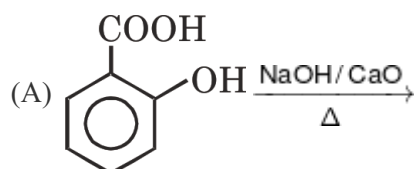
- (A) E°_{cell} is 0.43 V
- (B) Electrons flow from left to right electrode
- (C) The EMF of above cell at respective given concentrations $V^{3+}(0.1M)$, $VO^{3+}(0.01M)$, $H^+(1M)$, $Fe^{3+}(1M)$ and $Fe^{2+}(0.1M)$ is 0.52 volt.
- (D) If some VCl_3 is added to left half-cell, E_{cell} increases

Space for Rough Work

9. Identify reactions give ketone product ?



10. Which reaction can form phenol as one of the product



Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

1 litre of 1M CH_3COOH (very weak acid) taken in a container initially. Now this solution is diluted upto volume V (litre) so that pH of the resulting solution becomes the twice of the original value.

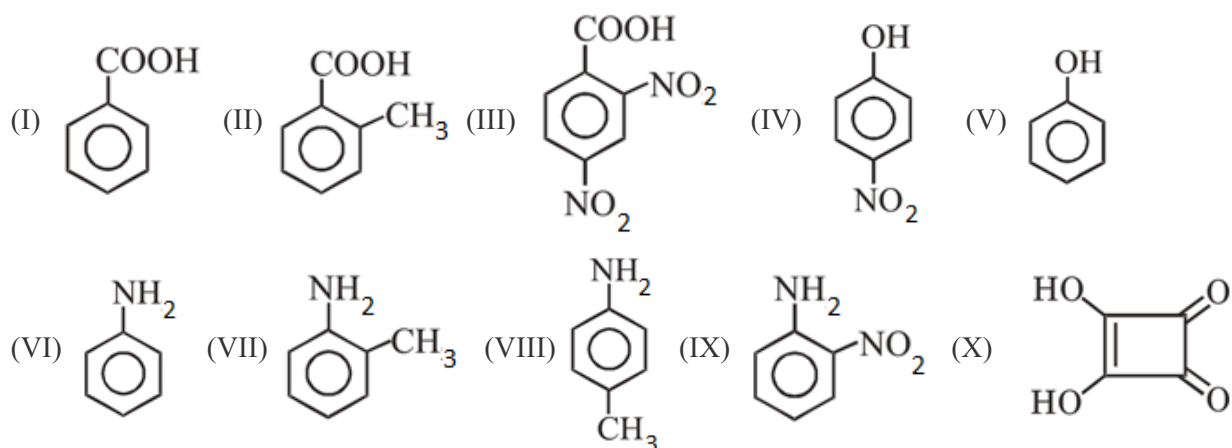
($K_a(\text{CH}_3\text{COOH}) = 10^{-6}$).

1. If $[\text{H}^+]$ in original solution is X^{-N} M then the value of $(X - N)$ is -
 2. If the value of V (in L) is $a \times 10^b$ then $(a + b)$ is :
-

Space for Rough Work

Paragraph for Questions 3 and 4

For the given compounds



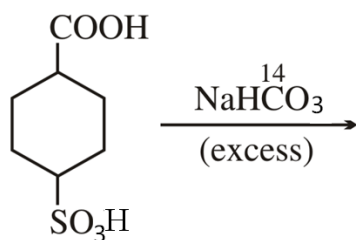
3. How many of the given compounds are soluble in NaOH :
4. How many compound give foul smell product when react with $\text{CHCl}_3 + \text{KOH}$?

Space for Rough Work

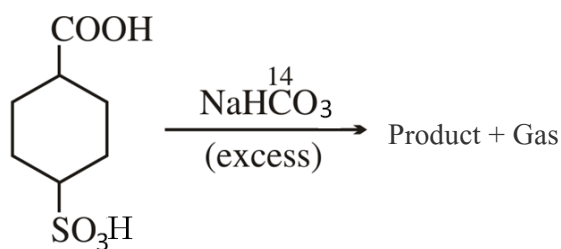
Paragraph for Questions 5 and 6

NaHCO_3 when treated with acids it may evolve CO_2 . Acids which are stronger & comparable acidic with H_2CO_3 may evolve CO_2 .

5. Molecular weight of evolved gas during reaction.



6. How many possible number of products (organic and inorganic) have equal C – O bond length.



Space for Rough Work

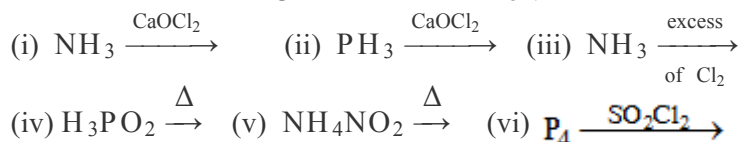
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

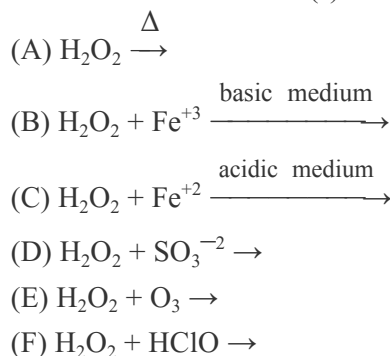
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Which of the following reaction form AX_3 (where $X = Cl / H$ & $A = N / P$)



8. Total number of reaction(s) in which oxygen is evolved.



9. For the just coagulation of 250 mL of $Fe(OH)_3$ sol, 2 ml of 1 M Na_2SO_4 electrolyte is required. What is the coagulating value of Na_2SO_4 electrolyte.

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. If $\hat{i} \times [(\vec{a} - \hat{j}) \times \hat{i}] + \hat{j} \times [(\vec{a} - \hat{k}) \times \hat{j}] + \hat{k} \times [(\vec{a} - \hat{i}) \times \hat{k}] = 0$ and let $\vec{b} = 2(\hat{i} + \hat{j} + \hat{k})$ then $(\vec{a} \cdot \vec{b})$ is
- (A) 1 (B) 2 (C) 3 (D) 4
2. If $z \in \mathbb{C}$ then area (in square unit) enclosed by the curve $2|z|^2 + |z|^2 + 3|z|^2 - |z|^2 = 6|z|$, $z \neq 0$ is
(Where $\mathbb{C}$ denotes set of complex number)
- (A) 1 (B) 2 (C) 3 (D) 6
-

Space for Rough Work

3. A box contains 2 fifty paise coins, 5 twenty-five paise coins and a certain number N (≥ 2) of ten and five paise coins. Five coins are taken out of the box at random. The probability that the total value of these coins is less than one rupee and fifty paise is

(A) $\frac{10(N+2)}{(N+7)C_5}$

(B) $1 - \frac{10(N+2)}{(N+7)C_5}$

(C) $1 - \frac{5(N+2)}{(N+7)C_5}$

(D) None of these

4. A box contains coupons labelled 1, 2, 3, 4, ..., 20. Five coupons are picked at random one after another without replacement. Let the coupons be x_1, x_2, x_3, x_4, x_5 in order. Then probability that $x_1 > x_2 > x_3 < x_4 < x_5$ is:

(A) $\frac{1}{20}$

(B) $\frac{1}{120}$

(C) $\frac{1}{24}$

(D) $\frac{1}{30}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ be vertices (in anti-clockwise order) of a trapezium ($AB \parallel CD$) in an argand plane. Let $|z_1 - z_2| = 4$, $|z_3 - z_4| = 10$ and the diagonals AC and BD intersect at P. It is given that $\arg\left(\frac{z_4 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}$ and $\arg\left(\frac{z_3 - z_2}{z_4 - z_1}\right) = -\frac{\pi}{4}$ then
- (A) Area of Trapezium is $\frac{140}{3}$ (B) $|CP - DP|$ is $\frac{10}{\sqrt{21}}$
- (C) area of ΔPCB is $\frac{400}{21}$ (D) Quadrilateral ABCD is concyclic
6. Let complex numbers Z_1 and $\frac{1}{Z_1}$ lie on $|Z - Z_0| = 4R$ and $|Z - Z_0| = 16R$ respectively. If Z_0 satisfies $2|Z_0|^2 = R^2 + 2$ and $|Z_1|^2 = \frac{p}{q}$ where $p, q \in \mathbb{N}$ and H.C.F of p and q is 1, then choose the correct option(s)
- (A) $p > q$ (B) $p < q$ (C) p is a prime (D) q is a prime

Space for Rough Work

7. Let the unit vectors $\vec{a}$ and $\vec{b}$ be perpendicular and the unit vector $\vec{c}$ be inclined at an angle θ to both $\vec{a}$ and $\vec{b}$. If $\vec{c} = \alpha\vec{a} + \beta\vec{b} + \gamma(\vec{a} \times \vec{b})$, (α, β, γ are scalars) then
- (A) $\alpha = \beta = \cos \theta$ (B) $\alpha = \cos \theta, \beta = \sin \theta$
 (C) $\gamma^2 = \cos 2\theta$ (D) $\gamma^2 = -\cos 2\theta$
8. If $2P(A \cup B) = 3P(A) = 4P(B) = 6P\left(\frac{A}{B}\right)$, then (where $P(x)$ denotes probability of occurrence of event x)
- (A) $P(A \cap \bar{B}) = \frac{1}{2}$ (B) $P\left(\frac{B}{A}\right) = \frac{1}{4}$
 (C) A & B are independent events (D) A & B are exhaustive events
9. Two identical tetrahedral dice with faces 1, 2, 3 and 4 shows even number with probability $\frac{1}{2}$ and $\frac{1}{3}$, respectively. If a_i and b_i are the integer generated by respective dice in i^{th} throws, and the probability that $a_1b_1 + a_2b_2 + \dots + a_kb_k$ is even is P_k . Then :
- (A) $P_4 > \frac{1}{2}$ (B) $P_5 < \frac{1}{2}$ (C) $P_2 = \frac{5}{9}$ (D) $3P_7 = 1 + P_6$
10. If $\vec{x} \times \vec{y} = \vec{a}$, $\vec{y} \times \vec{z} = \vec{b}$, $\vec{x} \cdot \vec{b} = \gamma$, $\vec{x} \cdot \vec{y} = 1$ & $\vec{y} \cdot \vec{z} = 1$ then ($|\vec{y}| \gamma \neq 0$)
- (A) $\vec{y} = \frac{\vec{a} \times \vec{b}}{\gamma}$ (B) $\vec{x} = \frac{1}{|\vec{y}|^2} [\vec{y} - \vec{a} \times \vec{y}]$
 (C) $\vec{z} = \frac{1}{|\vec{y}|^2} [\vec{y} + \vec{b} \times \vec{y}]$ (D) $\vec{x} = \frac{1}{|\vec{y}|^2} [\vec{y} + \vec{a} \times \vec{y}]$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

If the projection of the line $\frac{x}{2} = \frac{y-1}{2} = \frac{z-1}{1}$ on a plane P is $\frac{x}{1} = \frac{y-1}{1} = \frac{z-1}{-1}$. Then

1. Equation of plane P is $x + y + 2z = k$; where k is equal to
2. Square of distance of plane P from origin is

Paragraph for Questions 3 and 4

From a point P representing complex number z_1 on the curve $|z| = 2$ two tangents are drawn from P to the curve $|z| = 1$ meeting it at $A(z_2)$ and $B(z_3)$ then

3. The smallest positive value of $\arg\left(\frac{z_2}{z_3}\right)$ is (in degree)
4. The value of $\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right) \left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right)$

Paragraph for Questions 5 and 6

If the equation $|z|(z+1)^8 = |z+1| \cdot z^8$ (where $z \in \mathbb{C}$ and $z(z+1) \neq 0$) has distinct roots $z_1, z_2, \dots, z_n$ (where $n \in \mathbb{N}$), then

5. The absolute value of $\sum_{r=1}^n \operatorname{Re}(z_r)$ is equal to
6. The value of $\sum_{r=1}^n \operatorname{Im}(z_r)$ is equal to

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. Let $h(x) = (x^2 + x + 1)^{-1}$. The 36th derivative of $h(x)$ with respect to x computed at $x = 0$ is $k!$, then number of positive divisors of k is equal to
 8. Two players A and B play a series of matches. In each match it is 2 to 1 in favour of the winner of the previous match. If A wins the first match, then the probability that A will win 3 matches of the next 4 matches, is p where $81p$ is equal to - (Assume no match ends in draw)
 9. A slip of paper is given to a person A who marks it either with a plus sign or a minus sign. The probability of his writing a plus sign is $1/3$. A passes the slip to B, B passes it to C, C passes it to D and finally D passes it to a referee. Each of B,C,D either leave it alone or change the sign before passing it. B, C, D each change the sign with probability $2/3$. If the referee observes a plus sign on the slip then the probability that A originally wrote a plus sign is p , where $41p$ is equal to
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work